Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie / Lefapha la Boetšenere, Tikologo ya Kago le Theknolotši ya Tshedimošo

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Unit 4: Production Systems

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh

Quick Recap/Questions arising from Previous Lecture

The previous lecture focused on:

- Being able to define what a value chain is and linking this with the SIPOC approach, Porter's value chain and its application in businesses with respect to the primary and support activities, the red and blue ocean concepts and finally, an effort to link all of these to the semester project.

Expected Learning Outcomes for the Current Lecture

At the end of this lecture, you should:

- **Understand** the specific roles of a production manager in the setting up of a production department in a new business, towards the actualization of a set of systemic goals.
- **Understand** some vital basics about a production system from an applicative point of view, its significance in the design and development process of systems.
- **Understand some** basics about the Design and operation of a production system

Expected Learning Outcomes for the Current Lecture

- **Understand some vital attributes of a Production System such as:** Pull vs. Push production systems; Unit, Mass, Batch, Job shop, Continuous, Intermittent, heterogenous and homogenous production techniques. Production planning

- **Understand** vital processes and activities in a production system and finally,

- Be able to **apply** knowledge of a production system in the semester project

Presentation Outline

- Definition and Categorisation of a Production Systems into products and/or service systems
- Some Attributes of a Production Manager in Setting Up a Production Department/Portfolio
- Production System in the context of Systems Design and Development

Attributes of a Production System

- Pull vs. Push Production Systems
- Batch, Mass, and Continuous production/service techniques
- Processes & Activities in a Production System
- Production System in the Context of the Semester Project

Definition and Categorisation of a Production System [as component of a business]

- It's a system that is concerned with the efficient and effective creation of goods and/or rendering of services to the satisfaction of its client base, at an affordable cost and reliability.

- **Product-Oriented Production Systems**
  Creates physical products

- **Service-Oriented Production Systems**
  Render physical or virtual services

Examples of Product Oriented Systems

- Manufacturing
- Processing
- Building
- Carpentry
- Computer systems
- Robots
- Software development amongst others

Examples of Service Oriented Systems

- Consulting
- Communication
- Medical
- Policing
- Distributing
- Cleaning
- Hospitality
- Software applications amongst others

Hybrid Production Systems

- Manufacturing systems with full after-sales applicative services
- Software development and applications
- Military engineering and defense systems
- etc

Some Characteristics/Attributes of a Production Manager in Setting Up a Production Department

- Participate in the facilities Layout of the production shop floor
- Recommend Raw materials options, including production machines, etc.
- Design Machine
- Design Product/s
- Manufacture Product/s
- Recruit or Recommend Departmental Staff
- Plan for production

Production System Characteristics

- **PART B Section 1: Description of system characteristics based on its functional capabilities**
  - Utilise the developed system functions to address this section of the project.

Production System in the context of Systems Design and Development

The Production Phase comes after the design and development phases i.e. after (Conceptual and Engineering design phases).

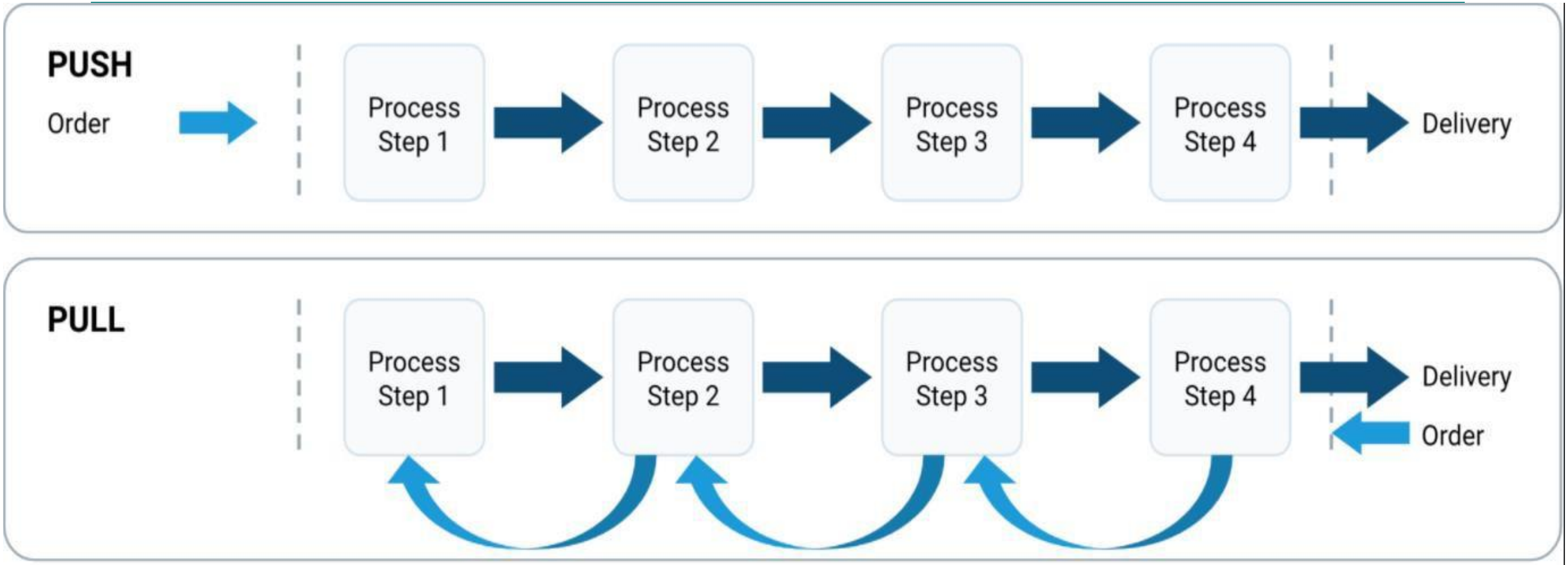
The Production Phase is a step away from the experimental labs/workshops/demonstration room or studio etc. It's a phase where products and/or services are meant to be created for deployment to the real world.

Generalised Attributes of a Production System

“Six Ms” of a Production System

- Man (humans generally i.e., women inclusive)
- Machine
- Material
- Method
- Money
- Management

Pull vs Push Production System



https://tulip.co/blog/what-is-a-push-system-vs-a-pull-system/

Pull Production System

- In a pull-based production system, distribution is often demand-driven rather than based on predictions. Goods are produced in the amount and time needed (Just in Time [JIT]).
- Pull systems are often preferred in situations where there is limited demand.
- They are also preferred when the cost of managing excess inventory outweighs the benefit of having a surplus of product in stock.

https://tulip.co/blog/what-is-a-push-system-vs-a-pull-system/

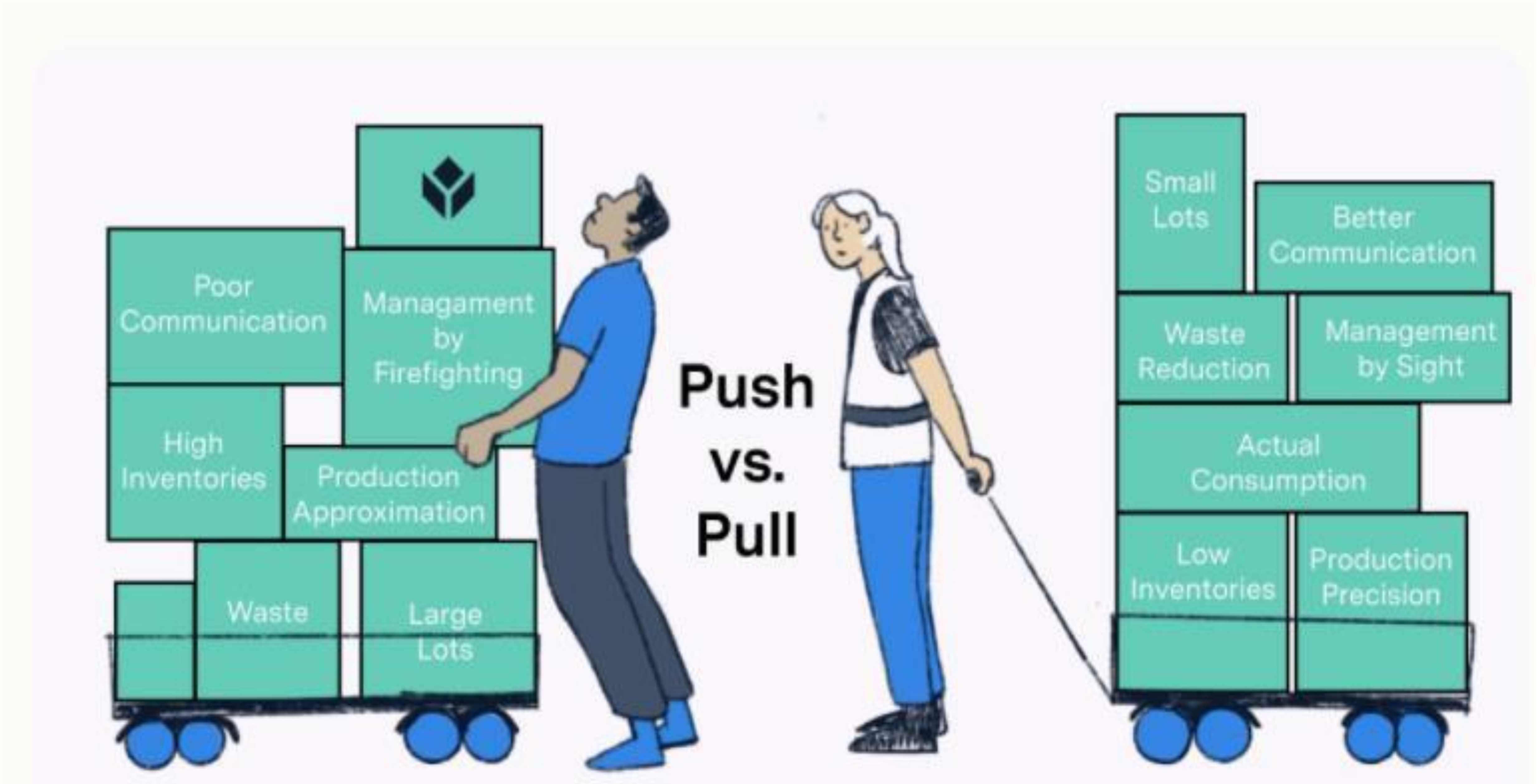
Push Production System

- In a push-based production system, products/service rendering are pushed through the channel from production up to the clients. This means that production happens based on demand forecast.

- Using a push system is preferable in instances where there is a high demand for a given product.

- Also, when having large amounts of inventory in stock is beneficial for meeting customer demand, a push system is preferred.

https://tulip.co/blog/what-is-a-push-system-vs-a-pull-system/



https://tulip.co/blog/what-is-a-push-system-vs-a-pull-system/

Unit, Mass, Batch, Continuous & Intermittent Production/Service Techniques

Mass Production/Services

Mass production is the processing of a large number of items at the same time. It is used to produce large units of a product on a continuum or continuous order e.g. production of cookies, muffins, automobiles, electronics etc or rendering of multi-services etc. **Even though Homogeneity of product type is expected, Heterogeneity** can play out here also.

Mass Production/Services

Unit production is the processing of a single unit piece of an item at any given time. It is used to produce large sized or complex products e.g. an aircraft such as an air bus.

Batch Production/Services

Batch production is a form of mass production with a defined number of items per run time. It is commonly used to produce several units of a product with some form of flexibility at different production periods & time e.g. different specs. or variants of the same product such as laptops, automobile, cookies, muffins etc or the mass rendering of the same type of service with varying add-ons e.g. dry cleaning services, etc.

In the batch production process, **homogeneity of practice** is the focus per batch.

Batch Techniques

Job shop Production/Services

A job shop production/service system is one that aims largely at product customization. Several specialised machines are available for diverse customization tasks. This also fits well into a unit production system

Some good examples include:

- Repair services
- Tool and die making
- Metal fabrication
- Prototyping

Continuous Production/Services

A continuous production system runs either manually or autonomously without a break, until the end of the planned production exercise. This can be applied to both mass or unit production systems or practices.

Intermittent Production/Services

An intermittent or discrete production system completes one production unit or more, stops, and resumes a new set after retooling, etc.

Some Core Processes in a Production System

- Production Organisation *(Setup & Planning/Preparations)*
- Production Operations *(Actual act of material processing or service rendering)*
- Production Standards Adherence *(Quality checks)*
- Production Packaging & Labelling *(Outbound finished products)*

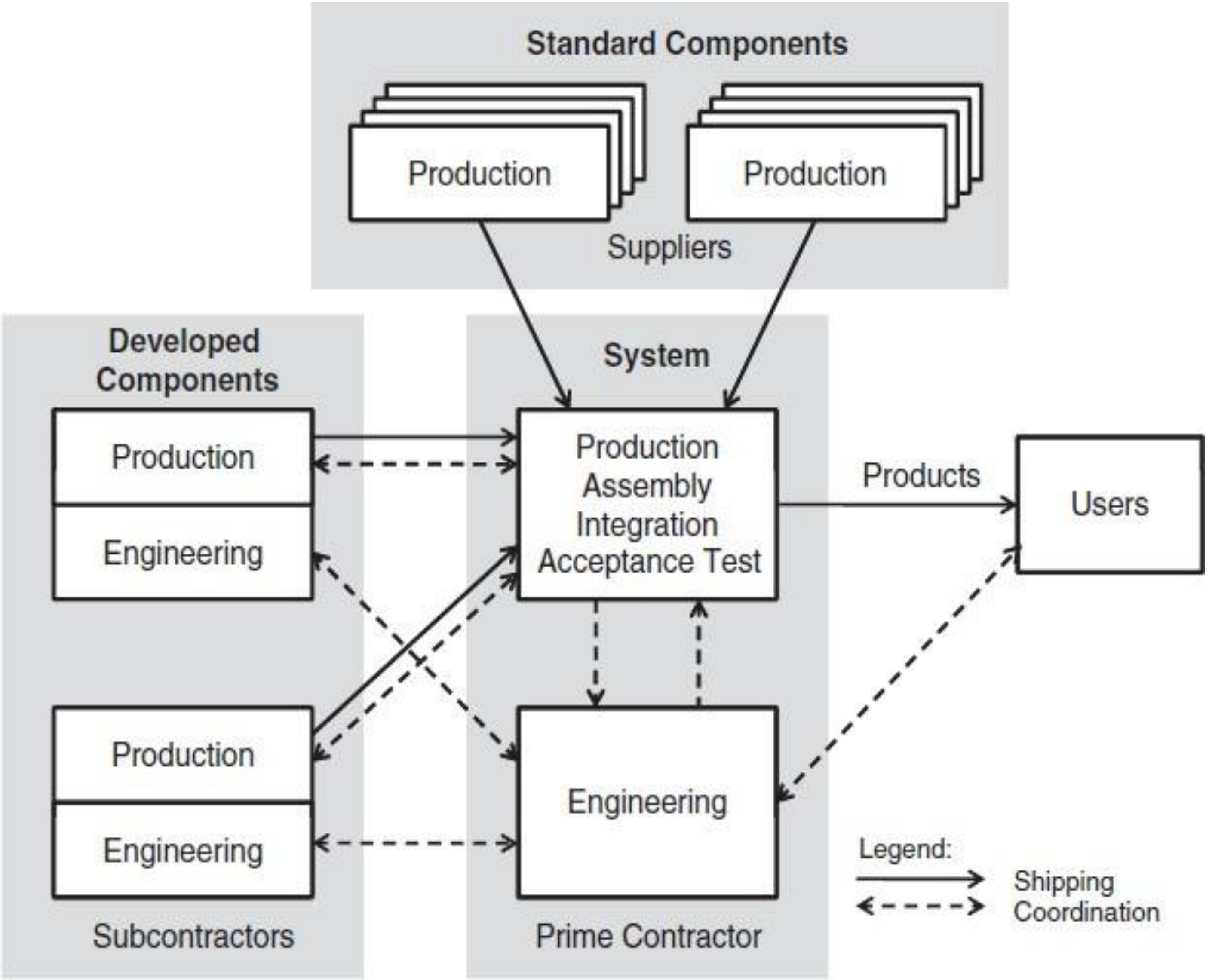
Some Common Activities in Production Processes

Tasks/Activities: Some vital things to do or put in place in order to have a successful Production System

- manufacturing sites and facilities
- tooling requirements, including special tools
- factory test equipment
- updating with engineering releases
- component fabrication
- components and parts inspection

Activities in a Production System…Contd

- production monitoring and control assembly
- acceptance test
- packaging, sealing and shipping
- discrepancy reports
- schedule and cost reports and
- production readiness reviews
- handling vs conveying



Make or Buy Tasks in a Production System

Production System in the Context of the Semester Project

For further reading: Sections covered in the e-book include:

Systems Engineering Principles and Practice, Second Edition; Author: Alexander Kossiakoff et al.

- Part One
  - Chapters 4: [4.2 (Production Phase)
- Part Four
  - Chapter 14: [14.4 and 14.5]

Other resources were randomly sourced:

The End

Next lecture

Unit 5: Logistics Systems

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